

Are the formulae satisfiable

$$(x_1 \vee x_2) \wedge (\bar{x}_1 \vee \bar{x}_2)$$

This is satisfiable for the following values $\{x_1 = 1, x_2 = 0\}$ or $\{x_1 = 0, x_2 = 1\}$

$$(x_1 \vee x_2) \wedge (x_1 \vee \bar{x}_2) \wedge (\bar{x}_1 \vee x_2) \wedge (\bar{x}_1 \vee \bar{x}_3)$$

Lets create a small truth table:

x_1	x_2	x_3	$x_1 \vee x_2$	$x_1 \vee \neg x_2$	$\neg x_1 \vee x_2$	$\neg x_1 \vee \neg x_3$
0	0	0	0	1	1	1
0	0	1	0	1	1	1
0	1	0	1	0	1	1
0	1	1	1	0	1	1
1	0	0	1	1	0	1
1	0	1	1	1	0	0
1	1	0	1	1	1	1
1	1	1	1	1	1	0

We can see that in the next to last line, $x_1 = 1, x_2 = 1, x_3 = 0$, we have a satisfiable solution

$$(x_1 \vee \bar{x}_2) \wedge (\bar{x}_1 \vee x_2 \vee x_3) \wedge (x_1 \vee \bar{x}_3)$$

Again, let's create a truth table:

x_1	x_2	x_3	$x_1 \vee \neg x_2$	$\neg x_1 \vee x_2 \vee x_3$	$x_1 \vee \neg x_3$
0	0	0	1	1	1
0	0	1	1	1	0
0	1	0	0	1	1
0	1	1	0	1	0
1	0	0	1	0	1
1	0	1	1	1	1
1	1	0	1	1	1
1	1	1	1	1	1

Here we have four satisfiable solutions, the first and also the three last

$$\{x_1, x_2, x_3\} = \begin{cases} \{0, 0, 0\} \\ \{1, 0, 1\} \\ \{1, 1, 0\} \\ \{1, 1, 1\} \end{cases}$$

Show that P is closed under union, concatenation and complement

Union

Since both languages are in P we know that they will run in polynomial time. Let's construct a turing machine that will decide on the union of two languages L_1 and L_2 , we call this machine $M_{\cup}$. $M_{\cup}$ functions as follows:

1. On some input i we run M_1 (turing machine for L_1) on input i
2. If it accepts, we accept
3. If it rejects, we run M_2 (turing machine for L_2) on input i
4. If it accepts, we accept
5. If it rejects, we reject

Concatenation

We can show this in a similar fashion as for union, here we construct a turing machine M_{+} that decides on the concatenation of two languages L_1 and L_2 that are both in P , now M_{+} functions as follows:

1. On some input i we run M_1 (turing machine for L_1) on input i
2. If M_1 rejects, we reject
3. If M_1 accepts, we run M_2 (turing machine for L_2) on input i
4. If M_2 accepts, we accept otherwise we reject

Complement

This is the easiest to show, now we have only on language L_1 that is in P we create a turing machine M_{-} that decides on the complement of L_1 , M_{-} functions as follows:

1. On some input i we run M_1 (turing machine for L_1) on input i
2. If M_1 accepts we reject
3. If M_1 rejects we accept